Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0–1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 1 · pretrain 8 · tie 0 → PRETRAIN

	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0–1)
ViT-g·2.0 pretrain_2X encoder	+11.62 [+9.26, +13.92]	**+0.22** [+0.22, +0.23]	-0.02 [-0.04, 0]	-0.14 [-0.14, -0.13]	0 [0, 0]	-0.12 [-0.12, -0.12]	0 [0, 0]	-0.01 [-0.01, -0.01]	-0.02 [-0.02, -0.02]	0.31
ViT-g·2.0 pretrain encoder	+11.67 [+9.48, +13.92]	+0.11 [+0.10, +0.11]	-0.01 [-0.03, +0.01]	-0.11 [-0.11, -0.11]	**0** [0, 0]	-0.09 [-0.09, -0.09]	**0** [0, 0]	0 [0, 0]	-0.01 [-0.01, -0.01]	0.65
ViT-g·2.0 pretrain head	+0.60 [-1.59, +2.74]	0 [0, 0]	**0** [-0.01, +0.02]	**0** [0, 0]	0 [+nan, +nan]	**0** [+nan, +nan]	0 [+nan, +nan]	**0** [+nan, +nan]	**0** [+nan, +nan]	0.62
ViT-g·2.0 surgery 3stage_DI encoder	+10.90 [+8.66, +13.21]	+0.11 [+0.11, +0.12]	-0.01 [-0.02, +0.01]	-0.07 [-0.08, -0.07]	0 [0, 0]	-0.07 [-0.07, -0.06]	0 [0, 0]	-0.02 [-0.02, -0.02]	-0.01 [-0.01, -0.01]	0.44
ViT-g·2.0 surgery 3stage_DI head	**+11.89** [+9.70, +14.19]	+0.11 [+0.11, +0.12]	-0.01 [-0.03, +0.01]	-0.11 [-0.11, -0.11]						0.57
ViT-g·2.0 surgery noDI encoder	+11.29 [+8.99, +13.59]	+0.12 [+0.11, +0.12]	-0.01 [-0.03, 0]	-0.07 [-0.07, -0.07]	0 [0, 0]	-0.06 [-0.07, -0.06]	0 [0, 0]	-0.02 [-0.02, -0.02]	-0.01 [-0.01, -0.01]	0.44
ViT-g·2.0 surgery noDI head	+10.36 [+8.00, +12.77]	+0.12 [+0.12, +0.12]	-0.01 [-0.02, +0.01]	-0.11 [-0.11, -0.11]						0.55
WINNER	S	P	P	P	P	P	P	P	P	